# HDOC Day-74

Implementation of Enumerations for Months and Operation Status in C

## Aim

To implement two C programs using enumerations, string comparisons, and switch-case statements: one to display the number of days in a selected month, and another to display a message for an operation status.

## Algorithm - Program 1: Months

1. Define enum Month with JAN through DEC.
2. Read a month abbreviation from the user.
3. Use strcmp() to identify the corresponding enumeration value.
4. Use switch to print 31 days, 30 days, or 28 or 29 days for February.
5. Print an error for an invalid month and terminate.

## Algorithm - Program 2: Operation Status

1. Define enum Status with SUCCESS, FAILURE, and TIMEOUT.
2. Read a status string and map it to an enumeration value using strcmp().
3. Use switch to display Operation Successful, Operation Failed, or Operation Timed Out.
4. Print an error for invalid input and terminate.

## C Program - Program 1: Months

```
#include <stdio.h>
#include <string.h>

enum Month {
    JAN, FEB, MAR, APR, MAY, JUN,
    JUL, AUG, SEP, OCT, NOV, DEC
};

int main()
{
    enum Month month;
    char input[20];
    printf("Enter month (JAN/FEB/MAR/.../DEC): ");
    scanf("%19s", input);

    if (strcmp(input, "JAN") == 0) month = JAN;
    else if (strcmp(input, "FEB") == 0) month = FEB;
    else if (strcmp(input, "MAR") == 0) month = MAR;
    else if (strcmp(input, "APR") == 0) month = APR;
    else if (strcmp(input, "MAY") == 0) month = MAY;
    else if (strcmp(input, "JUN") == 0) month = JUN;
    else if (strcmp(input, "JUL") == 0) month = JUL;
    else if (strcmp(input, "AUG") == 0) month = AUG;
    else if (strcmp(input, "SEP") == 0) month = SEP;
    else if (strcmp(input, "OCT") == 0) month = OCT;
    else if (strcmp(input, "NOV") == 0) month = NOV;
    else if (strcmp(input, "DEC") == 0) month = DEC;
    else {
        printf("Invalid month!\n");
        return 0;
    }

    switch (month) {
        case JAN: case MAR: case MAY: case JUL:
        case AUG: case OCT: case DEC:
            printf("31 days\n"); break;
        case APR: case JUN: case SEP: case NOV:
            printf("30 days\n"); break;
        case FEB:
            printf("28 or 29 days\n"); break;
        default:
            printf("Invalid month!\n");
    }
    return 0;
}
```

## C Program - Program 2: Operation Status

```
#include <stdio.h>
#include <string.h>

enum Status { SUCCESS, FAILURE, TIMEOUT };

int main()
{
    enum Status status;
    char input[20];
    printf("Enter status (SUCCESS/FAILURE/TIMEOUT): ");
    scanf("%19s", input);

    if (strcmp(input, "SUCCESS") == 0) status = SUCCESS;
    else if (strcmp(input, "FAILURE") == 0) status = FAILURE;
    else if (strcmp(input, "TIMEOUT") == 0) status = TIMEOUT;
    else {
        printf("Invalid operation status!\n");
        return 0;
    }

    switch (status) {
        case SUCCESS:
            printf("Operation Successful\n"); break;
        case FAILURE:
            printf("Operation Failed\n"); break;
        case TIMEOUT:
            printf("Operation Timed Out\n"); break;
        default:
            printf("Invalid operation status!\n");
    }
    return 0;
}
```

## Explanation

An enumeration (enum) defines a set of named integer constants. In enum Month, JAN starts at 0 and DEC has value 11. In enum Status, SUCCESS = 0, FAILURE = 1, and TIMEOUT = 2. The programs compare text inputs using strcmp(), then use switch-case to select the appropriate output.

Months with 31 days: JAN, MAR, MAY, JUL, AUG, OCT, DEC. Months with 30 days: APR, JUN, SEP, NOV. FEB has 28 or 29 days depending on the year; because no year is requested, the program prints both possibilities.

## Test Case Table

| Case | Program | Input | Expected Output |
|---|---|---|---|
| 1 | Months | JAN | 31 days |
| 2 | Months | FEB | 28 or 29 days |
| 3 | Months | APR | 30 days |
| 4 | Months | DEC | 31 days |
| 5 | Months | ABC | Invalid month! |
| 6 | Status | SUCCESS | Operation Successful |
| 7 | Status | FAILURE | Operation Failed |
| 8 | Status | TIMEOUT | Operation Timed Out |
| 9 | Status | UNKNOWN | Invalid operation status! |

## Sample Outputs

### February

```
Enter month (JAN/FEB/MAR/.../DEC): FEB
28 or 29 days
```

### April

```
Enter month (JAN/FEB/MAR/.../DEC): APR
30 days
```

### Successful Operation

```
Enter status (SUCCESS/FAILURE/TIMEOUT): SUCCESS
Operation Successful
```

### Failed Operation

```
Enter status (SUCCESS/FAILURE/TIMEOUT): FAILURE
Operation Failed
```

### Timeout

```
Enter status (SUCCESS/FAILURE/TIMEOUT): TIMEOUT
Operation Timed Out
```

### Invalid Month

```
Enter month (JAN/FEB/MAR/.../DEC): ABC
Invalid month!
```

## Result

The programs demonstrate enumerations, user-defined types, string comparison, and switch-case statements. The first displays the days in a month, and the second displays the message corresponding to SUCCESS, FAILURE, or TIMEOUT.